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Cluster dynamics modelling of materials: A new hybrid deterministic/stochastic coupling approach

Pierre Terrier [a,b], Manuel Athènes [b], Thomas Jourdan [b,*], Gilles Adjanor [c], Gabriel Stoltz [a]

[a] Université Paris-Est, CERMICS (ENPC), INRIA, F-77455 Marne-la-Vallée, France
[b] DEN-Service de Recherches de Métallurgie Physique, CEA, Université Paris-Saclay, F-91191 Gif-sur-Yvette, France
[c] EDF R&D, Département Matériaux et Mécanique des Composants, Les Renardières, F-77250 Moret sur Loing, France

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Deterministic simulations of the rate equations governing cluster dynamics in materials are limited by the number of equations to integrate. Stochastic simulations are limited by the high frequency of certain events. We propose a coupling method combining deterministic and stochastic approaches. It allows handling different time scale phenomena for cluster dynamics. This method, based on a splitting of the dynamics, is generic and we highlight two different hybrid deterministic/stochastic methods. These coupling schemes are highly parallelizable and specifically designed to treat large size cluster problems. The proof of concept is made on a simple model of vacancy clustering under thermal ageing.

1. Introduction

The microstructural evolution of materials under thermal ageing or irradiation involves complex processes, such as nucleation, growth and coarsening of precipitates or bubbles, that occur on different time scales. The computer simulation of such processes triggered the development of efficient methods able to deal with very different time scales [1–17]. Long time scale phenomena arise in the evolution of the microstructure of materials under thermal ageing or irradiation. To simulate such events (nucleation, formation of precipitates, growth of bubbles, etc.) one needs efficient methods that are able to handle systems with different time scales. Monte Carlo methods, such as kinetic Monte Carlo [1–3], give physically accurate results but may be limited to short time simulations when frequent events occur.

Mean-field techniques such as rate equation cluster dynamics (RECD) have been used with success to get around this issue [4–6]. The modelling of the microstructure is approximated by considering only the defect concentrations, whose evolutions are determined by a system of ordinary differential equations (ODE), called rate equations. Despite its simplicity, two main difficulties occur with RECD. First, since there is one rate equation per cluster type, the number of equations might become very large (clusters might contain up to millions of atoms or defects). In addition, such systems of ODEs are generally stiff, *i.e.* the typical time scale for some reactions is very large while it may be very small for others. Implicit methods are then required and solving such a system of ODE becomes computationally prohibitive as the cluster sizes increase.

* Corresponding author.
E-mail address: thomas.jourdan@cea.fr (T. Jourdan).

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Several methods and approximations have been proposed to solve these equations. Deterministic ones include grouping methods where rate equations are gathered into classes [7,8], and a Fokker–Planck approach where rate equations for large size clusters are approximated by a Fokker–Planck equation [9]. Recent developments of the Fokker–Planck approach [6,10] have proven to be really efficient when only one or two types of defect are considered. They are however strongly limited by the dimensionality of the system. To our knowledge no system with three types of defects/solutes or more have been simulated using a deterministic approach up to large size clusters.

Marian et al. recently proposed a stochastic implementation of the rate theory cluster dynamics [13]. This method is intended to take into account complex clusters containing different species (point defects, atoms, etc.). The formalism of RECD has been related to purely stochastic approaches such as the well known Stochastic Simulation Algorithm (SSA) introduced by Gillespie [12]. Nevertheless, in stiff systems where certain reactions occur frequently, the efficiency of the computations is still limited by discrepancies in the time scales.

Attempts have been made to take advantage of both deterministic and stochastic methods. Hybrid deterministic/stochastic algorithms have been proposed [15,16]. Rate equations are used for small size clusters, while large size ones are treated stochastically. In particular, Surh et al. [15] approximate the evolution of large size clusters with a Fokker–Planck equation and use a Langevin dynamics to propagate stochastic particles when clusters reach a certain size. While, to our knowledge, this method is the first one to use such a coupling for cluster dynamics, it has an important limitation. As stochastic particles representing clusters of a certain size are emitted one by one at each time step, the time step should be small to increase the number of particles, and hence reduce the statistical noise. On the contrary, if one wants to rapidly reach large time scales, the time step should be large.

In this work, we propose an alternative way to couple deterministic and stochastic simulations. After a brief presentation of the physical model, we introduce in Section 2 a first splitting between the vacancy concentration and the remainder of the distribution. Hence, when the vacancy concentration is fixed, the cluster dynamics becomes linear. Taking advantage of this linearity feature, the dynamics is further decomposed, this time between small and large size clusters. We describe a generic version of the algorithm based on these two splittings. This generic method allows us to design several coupling methods depending on the way the subdynamics are solved. In particular we introduce in Section 3 two stochastic methods for computing the evolution of large size clusters, one based on the discrete Markov process associated with the rate equations and the other on a Fokker–Planck approximation. In Section 4, we present different ways of computing the vacancy concentration. In Section 5 we present a numerical analysis of the algorithm. Numerical results are presented in Section 6. We compare in particular some of the methods to compute the vacancy concentration in order to confirm both the validity of our approximation and the overall accuracy of the method.

2. Model description and main algorithm

We study vacancy clustering during ageing with the model system described in [18]. The chosen model is simple but can be enriched with additional sink/source terms, mobile clusters of size two or greater, etc.

2.1. Rate equations

The RECD approach is used to describe the evolution of cluster size concentration $(C_{\mathrm{vac}}, C_2, C_3, \cdots)$ where C_{vac} is the vacancy concentration and C_n is the concentration of a cluster of size n, *i.e.* containing n vacancies. A set of rate equations governing the time evolution of each concentration is solved. We assume that only mono-vacancies are mobile. Therefore, the rate equation for the concentration C_n of an immobile cluster of size $n \geq 2$ is

$$\frac{dC_n}{dt} = \beta_{n-1} C_{n-1} C_{\mathrm{vac}} - (\beta_n C_{\mathrm{vac}} + \alpha_n) C_n + \alpha_{n+1} C_{n+1}, \tag{1}$$

where β_n is the absorption rate and α_n the emission rate. These rates take the form [18]

$$\begin{cases} \beta_n = \beta_0 n^{1/3}, \quad n \geq 1, \\ \alpha_n = \alpha_0 n^{1/3} \exp\left(-\dfrac{E^{\mathrm{b}}_{\mathrm{vac}}(n)}{k_{\mathrm{B}} T} \right), \end{cases} \tag{2}$$

where $\alpha_0 = \beta_0 = (48\pi^2/V_{\mathrm{at}}^2)^{1/3} D_{\mathrm{vac}}$, with V_{at} the atomic volume and D_{vac} the diffusion coefficient of vacancies. The term $E^{\mathrm{b}}_{\mathrm{vac}}(n)$ represents the binding energy of a vacancy with a cluster of size n:

$$E^{\mathrm{b}}_{\mathrm{vac}}(n) = E^{\mathrm{f}}_{\mathrm{vac}} - \frac{2\gamma V_{\mathrm{at}}}{r(n)}, \tag{3}$$

where γ is a surface energy, r is the radius of void given by $r(n) = (3nV_{\mathrm{at}}/4\pi)^{1/3}$ and $E^{\mathrm{f}}_{\mathrm{vac}}$ is the vacancy formation energy. The rate equation for C_{vac} is given by:

$$\frac{dC_{\mathrm{vac}}}{dt} = -2\beta_1 (C_{\mathrm{vac}})^2 - \sum_{n\geq 2} \beta_n C_n C_{\mathrm{vac}} + \sum_{n \geq 2} \alpha_n C_n + \alpha_2 C_2. \tag{4}$$

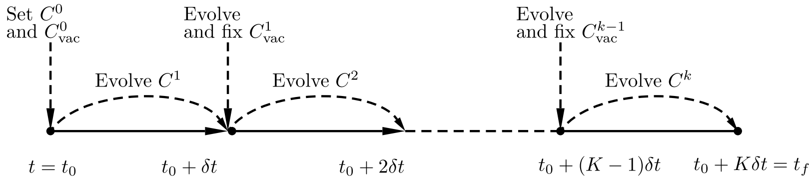

Fig. 1. Illustration of the splitting between the dynamics of C_{vac} and $(C_n)_{n\geq 2}$.

The latter equation is obtained by requiring that cluster dynamics preserve the total quantity of matter $\mathcal{Q}_{\mathrm{tot}}$:

$$\frac{d\mathcal{Q}_{\mathrm{tot}}}{dt} = \frac{d}{dt}\left(C_{\mathrm{vac}} + \sum_{n\geq 2} nC_n\right) = 0. \tag{5}$$

Initial conditions are given by

$$C_{\mathrm{vac}}(0) = C_{\mathrm{init}} \quad \text{and} \quad C_n(0) = 0, \quad n \geq 2, \tag{6}$$

where C_{init} is the quenched-in vacancy concentration.

2.2. Splitting of the dynamics

Solving the set of rate equations (1)–(4) when large clusters appear becomes computationally prohibitive. One way to address this problem is to numerically solve the evolution of different classes of clusters with dedicated methods [6,15]. We present here a generic algorithm that allows a seamless coupling between such methods.

We propose to first split the dynamics into two elementary dynamics, namely the dynamics of the vacancy concentration C_{vac} at fixed concentrations $(C_n)_{n\geq 2}$ (see Eq. (4)) and the dynamics of the cluster concentrations $C = (C_n)_{n\geq 2}$ at fixed vacancy concentration C_{vac} (see Eq. (1)). This splitting may be performed after a time t_0 corresponding to some initial transient regime where the full set of ODEs (1)–(4) is integrated by a standard numerical scheme. Let δt be a time step and $t_k = t_0 + k\delta t$ for $k \in \{1, \cdots, K\}$ the time at which approximations of the solution are sought, C^k_{vac} and C^k being respectively approximate solutions of $C_{\mathrm{vac}}(t_k)$ and $C(t_k)$. A good approximation of the cluster dynamics is provided by the following procedure:

$$\begin{cases} C^k = \mathcal{G}_{\delta t}\left(C^{k-1}; C^{k-1}_{\mathrm{vac}}\right), \\ C^k_{\mathrm{vac}} = \mathcal{F}_{\delta t}\left(C^{k-1}_{\mathrm{vac}}; C^k\right), \end{cases} \tag{P1}$$

where $\mathcal{F}_{\delta t}$ and $\mathcal{G}_{\delta t}$ respectively approximate the evolution of (4) and (1) over a time step δt. Fig. 1 illustrates such a splitting.

Notice that in the limit $\delta t \to 0$, the problem (P1) becomes equivalent to the full cluster dynamics (1)–(4). The error is determined by the time step δt and the quality of the approximations $\mathcal{F}_{\delta t}$ and $\mathcal{G}_{\delta t}$. We quantify the error with respect to δt in Section 6.1, where we also discuss the range of admissible δt. Since this splitting is a Lie–Trotter splitting, the error on finite time properties scales as δt when the sub-ODEs are integrated exactly (by an analysis similar to the one provided in [19]), as illustrated in Section 5.1. In practice, an additional error arises from the fact that $\mathcal{F}_{\delta t}$ and $\mathcal{G}_{\delta t}$ are not exact, and are typically constructed by substepping strategies; see Section 4.1 for $\mathcal{F}_{\delta t}$.

2.2.1. Integrating the vacancy subdynamics

The numerical scheme $\mathcal{F}_{\delta t}$ in (P1) is obtained by approximating the solution of Eq. (4) with $C = (C_n)_{n\geq 2}$ fixed. Let us now discuss how to obtain C^k_{vac} from the knowledge of C^{k-1}_{vac} and $C^k = (C^k_n)_{n\geq 2}$. Introducing

$$\mathcal{A}^k = \sum_{n\geq 2} \alpha_n C^k_n + \alpha_2 C^k_2 \tag{7}$$

and

$$\mathcal{B}^k = \sum_{n\geq 2} \beta_n C^k_n, \tag{8}$$

C^k_{vac} is an approximation of the solution of the following dynamics at time δt:

$$\frac{dC_{\mathrm{vac}}}{dt} = -\beta_1 \left(C_{\mathrm{vac}}(t)\right)^2 - \mathcal{B}^k C_{\mathrm{vac}}(t) + \mathcal{A}^k, \quad C_{\mathrm{vac}}(0) = C^{k-1}_{\mathrm{vac}}. \tag{9}$$

The actual numerical method $\mathcal{F}_{\delta t}$ depends on the numerical scheme used to integrate (9) (see Section 4).

2.2.2. Integrating the cluster subdynamics

The numerical scheme $\mathcal{G}_{\delta t}$ in (P1) is obtained by approximating the solution of Eq. (1) with C_{vac} fixed. Let us now discuss how to obtain C^k from the knowledge of C_{vac}^{k-1} and C^{k-1}. First notice that, when the vacancy concentration is fixed, the set of rate equations (1) forms a linear problem, that can be expressed in matrix form. Denote by $(e_n)_{n\geq 2}$ the basis with components $(e_n)_i = \delta_n(i)$, where δ is the Kronecker delta. Let A_0 be the tridiagonal operator such that:

$$
\begin{aligned}
&A_0(C_{\text{vac}})e_2 = -(\beta_2 C_{\text{vac}} + \alpha_2)e_2 + \beta_2 C_{\text{vac}} e_3, \\
&A_0(C_{\text{vac}})e_n = \alpha_n e_{n-1} - (\beta_n C_{\text{vac}} + \alpha_n)e_n + \beta_n C_{\text{vac}} e_{n+1}, \quad n \geq 3.
\end{aligned}
\tag{10}
$$

The operator A_0 can also be represented as the following infinite matrix:

$$
A_0(C_{\text{vac}}) = \begin{pmatrix}
-(\beta_2 C_{\text{vac}} + \alpha_2) & \alpha_3 & 0 & 0 & \cdots \\
\beta_2 C_{\text{vac}} & -(\beta_3 C_{\text{vac}} + \alpha_3) & \alpha_4 & 0 & \cdots \\
0 & \beta_3 C_{\text{vac}} & -(\beta_4 C_{\text{vac}} + \alpha_4) & \alpha_5 & \cdots \\
0 & 0 & \beta_4 C_{\text{vac}} & -(\beta_5 C_{\text{vac}} + \alpha_5) & \ddots \\
\vdots & \vdots & \vdots & \ddots & \ddots
\end{pmatrix}
\tag{11}
$$

The approximation C^k is obtained by numerically integrating the following dynamics:

$$
\begin{cases}
\dfrac{dC}{dt} = A_0\left(C_{\text{vac}}^{k-1}\right) C, \\
C(0) = C^{k-1},
\end{cases}
\tag{P2}
$$

depending on the numerical scheme used to integrate (P2).

As previously noticed, a key feature of Eq. (1) is that the dynamics is linear. For any initial condition C^0, it is then possible to split the evolution problem (P2) into independent evolutions, corresponding to a decomposition of the initial condition C^0. The solution is then obtained by summing the independent sub-solutions. If one writes $C^0 = C^{0,a} + C^{0,b}$, then the solution is $C(t) = C^a(t) + C^b(t)$ with $C^z(t)$ the solution of (P2) with initial condition $C^{0,z}$ for $z = a, b$.

2.2.3. Splitting and decomposition of the dynamics

Using the linearity of (P2) we choose to separate the evolution of small and large size clusters. With initial conditions C_{small}^{k-1} and C_{large}^{k-1} such that $C_{\text{small}}^{k-1} + C_{\text{large}}^{k-1} = C^{k-1}$, the main problem (P1) now writes:

$$
\begin{cases}
C_{\text{small}}^k = \mathcal{G}_{\delta t}^{\text{small}}\left(C_{\text{small}}^{k-1}; C_{\text{vac}}^{k-1}\right), \\
C_{\text{large}}^k = \mathcal{G}_{\delta t}^{\text{large}}\left(C_{\text{large}}^{k-1}; C_{\text{vac}}^{k-1}\right), \\
C_{\text{vac}}^k = \mathcal{F}_{\delta t}\left(C_{\text{vac}}^{k-1}; C_{\text{small}}^k + C_{\text{large}}^k\right).
\end{cases}
\tag{P3}
$$

Note that such a decomposition between small and large size clusters allows us to solve the corresponding dynamics with a different numerical scheme (as emphasized by the notations $\mathcal{G}^{\text{small}}$ and $\mathcal{G}^{\text{large}}$). It is straightforward and computationally effective to numerically solve rate equations for small size clusters (since they consist of a small number of ODEs), so that many options are available for $\mathcal{G}^{\text{small}}$. On the other hand, the treatment of large size clusters requires dedicated techniques. We present in Section 3 two stochastic methods that are highly parallelizable and more appropriate for large size clusters. Fig. 2 illustrates the decomposition between small and large size clusters on a single time interval. In Fig. 2(a), the initial distribution is divided into two distributions, one for small size clusters, the other for large size ones. Both distributions are then propagated independently over time and the sum of both propagated distributions (Fig. 2(b)) gives us an approximation of the total distribution.

2.3. Main algorithm

From the discussion of Section 2.2, the introduction of the coupling algorithm is rather straightforward. Let us introduce a final time t_f for the calculation, a frontier N_{front} and a buffer zone of size N_{buff} for the separation and overlapping of small and large size clusters. The size of the buffer zone is chosen sufficiently small to limit the computational cost, as it allows in particular to reduce the number of ODE to solve. Therefore one has to choose N_{buff} and δt such that when one propagates the distribution of small size clusters with $C_n = 0$ for $n \geq N_{\text{front}}$ on a time step δt, it remains negligible for $n \geq N_{\text{front}} + N_{\text{buff}}$. Actually, the distribution around n approximately propagates at an average speed of $\beta_n C_{\text{vac}} - \alpha_n$. This property can be observed easily on the Fokker–Planck equation (17), presented in Section 3.2, where the quantity

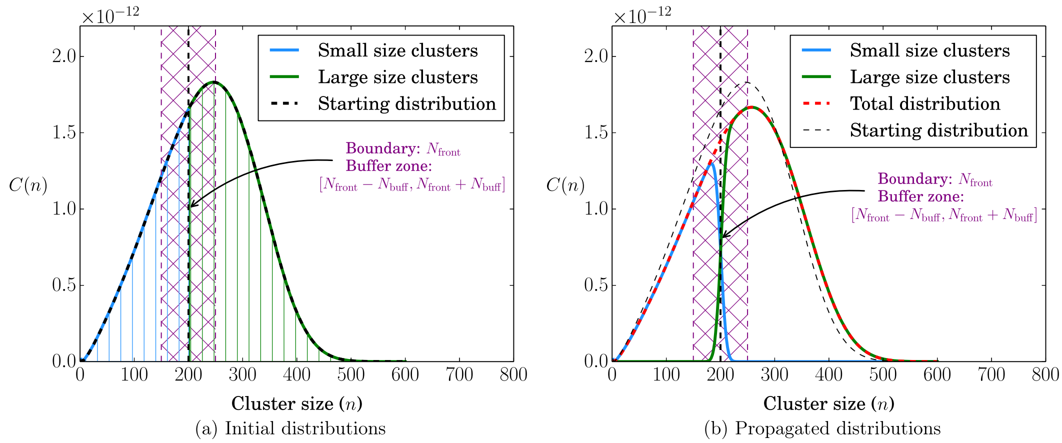

Fig. 2. Illustration of the decomposition of the linear dynamics: the initial distribution is divided into two distributions, which are independently propagated. The buffer zone allows the distributions to overlap and limits the calculation cost since both distributions are propagated on a limited space.

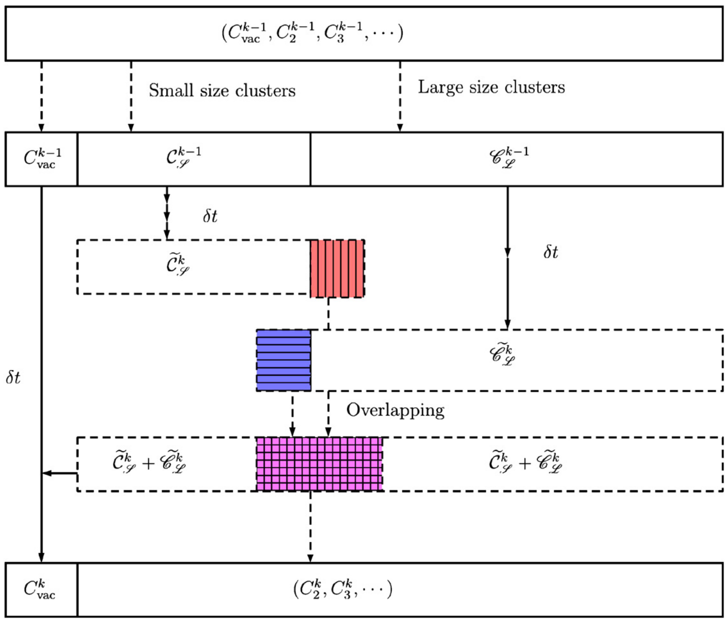

Fig. 3. Summary of the algorithm presented in Section 2.3.

$F \triangleq \beta C_{\text{vac}} - \alpha$ will act as a drift term. Therefore N_{buff} can be chosen to be of order $(\beta_{N_{\text{front}}} C_{\text{vac}} - \alpha_{N_{\text{front}}})\delta t$. We also set a maximum cluster size to N_{max}.

As mentioned in Section 2.2, we intend to solve the evolution of small and large size clusters with different methods. While the scheme $\mathcal{G}^{\text{small}}$ can be as simple as a Euler scheme for ODEs, the scheme $\mathcal{G}^{\text{large}}$ is more elaborated and will be detailed in Section 3. The diagram presented in Fig. 3 summarizes the algorithm presented hereafter.

Let $C^0 = (C^0_n)_{n\geq 2}$ be the initial distribution of the cluster concentrations. We denote by $\mathcal{C}^0_{\mathscr{S}} = (C^0_2, \cdots, C^0_{N_{\text{front}}-1}, 0, \cdots)$ the initial distribution for small size clusters and $\mathscr{C}^0_{\mathscr{L}} = (0, \cdots, 0, C^0_{N_{\text{front}}}, C^0_{N_{\text{front}}+1}, \cdots)$ the initial distribution for large size clusters and C^0_{vac} the initial vacancy concentration. To compute the solution from a time $k\delta t$ to a time $(k+1)\delta t$, the general algorithm reads as follows:

(0) Decompose the total distribution between small size and large size clusters:

$$\begin{aligned}
\mathcal{C}^k_{\mathscr{S}} &= \left(\widetilde{\mathcal{C}}^k_{\mathscr{S}}(2), \cdots, \widetilde{\mathcal{C}}^k_{\mathscr{S}}(N_{\text{front}}-1), 0, \cdots\right), \\
\mathscr{C}^k_{\mathscr{L}} &= \left(0, \cdots, 0, \widetilde{\mathscr{C}}^k_{\mathscr{L}}(N_{\text{front}}), \widetilde{\mathscr{C}}^k_{\mathscr{L}}(N_{\text{front}}+1), \cdots\right).
\end{aligned} \tag{M0}$$

(1) Compute $\widetilde{\mathcal{C}}_{\mathscr{S}}^{k+1}$ on $\{2,\cdots,N_{\mathrm{front}}+N_{\mathrm{buff}}\}$ by integrating the ODE (1) with initial condition $\mathcal{C}_{\mathscr{S}}^{k}$ equal to 0 for $n \geq N_{\mathrm{front}}$:

$$\widetilde{\mathcal{C}}_{\mathscr{S}}^{k+1} = \mathcal{G}_{\delta t}^{\mathrm{small}}\left(\mathcal{C}_{\mathscr{S}}^{k}; C_{\mathrm{vac}}^{k}\right). \tag{M1}$$

(2) Compute $\widetilde{\mathscr{C}}_{\mathscr{L}}^{k+1}$ on $\{N_{\mathrm{front}}-N_{\mathrm{buff}},\cdots,N_{\max}\}$ by a (possibly approximate) dynamics for large size clusters, with initial condition $\mathscr{C}_{\mathscr{L}}^{k}$ equal to 0 for $n \leq N_{\mathrm{front}}-1$:

$$\widetilde{\mathscr{C}}_{\mathscr{L}}^{k+1} = \mathcal{G}_{\delta t}^{\mathrm{large}}\left(\mathscr{C}_{\mathscr{L}}^{k}; C_{\mathrm{vac}}^{k}\right). \tag{M2}$$

(3) Compute the total distribution C^{k+1}:

$$C^{k+1} = \widetilde{\mathcal{C}}_{\mathscr{S}}^{k+1} + \widetilde{\mathscr{C}}_{\mathscr{L}}^{k+1}. \tag{M3}$$

(4) Update the vacancy concentration C_{vac}^{k+1}:

$$C_{\mathrm{vac}}^{k+1} = \mathcal{F}_{\delta t}\left(C_{\mathrm{vac}}^{k}; C^{k+1}\right). \tag{M4}$$

Let us make some general remarks:

- Before the distribution reaches the border N_{front}, equations (1) and (4) can be solved by an ODE scheme without any splitting between C_{vac} and $(C_n)_{n\geq 2}$. This may be important for the initial transient regime during which C_{vac} rapidly evolves.
- An alternative way to handle the boundary and the buffer zone is to introduce adaptive time steps δt for a fixed value N_{buff}. These steps would be limited by the requirement that $\mathcal{C}_{\mathscr{S}}(N_{\mathrm{front}}+N_{\mathrm{buff}})$ should not grow to a value that is not negligible any more.
- Since Steps (M1) and (M2) are independent, it is possible to switch those steps or to perform them simultaneously.

3. Discretization of the large size cluster subdynamics

As we want to avoid the computation of a large number of ODEs, we present two methods that allow for a stochastic and parallelizable solving of the evolution of large size clusters (Step (M2) in the algorithm of Section 2.3). The first method, called Jump process approach, relies on an original formulation of the set of rate equations when C_{vac} is fixed, and is better suited for clusters of intermediate sizes. The second method, called Langevin process approach, is based on an approximation by a partial differential equation of the set of rate equations for large size clusters, which is more appropriate for sufficiently large clusters.

3.1. The Jump process approach

3.1.1. Model presentation

Assuming that the vacancy concentration C_{vac} is fixed, the rate equations (1) are equivalent to a forward-Kolmogorov equation of a Markov process. Such an observation was already made by Goodrich [20]. Indeed a cluster of size n either emits a vacancy and reduces to a cluster of size $n-1$ or absorbs a vacancy and grows to a cluster of size $n+1$. In the size space, such a behaviour can be seen as the jump of a particle in state n to a state $n-1$ or to a state $n+1$. Since the rate of absorption depends on C_{vac}, the Markov process is actually a time-dependent Jump process. It is described as follows: consider a population of size $N(t)$ at a time t and a small time step $\delta\tau$. The transition probabilities for $n \geq 2$ are given by

$$\begin{aligned}
&\mathbb{P}[N(t+\delta\tau)=n+1|N(t)=n] = \beta_n C_{\mathrm{vac}}\delta\tau + \mathrm{O}(\delta\tau),\\
&\mathbb{P}[N(t+\delta\tau)=n-1|N(t)=n] = \alpha_n\delta\tau + \mathrm{O}(\delta\tau),\\
&\mathbb{P}[N(t+\delta\tau)=n|N(t)=n] = 1-(\beta_n C_{\mathrm{vac}}+\alpha_n)\delta\tau + \mathrm{O}(\delta\tau).
\end{aligned} \tag{12}$$

Let $p(t,n)=\mathbb{P}(N(t)=n)$ be the probability to be in state n at a time t. The evolution of $(p(t,n))_{n\geq 2}$ is then governed by the following dynamics, a forward-Kolmogorov equation:

$$\frac{dp}{dt}(t,n) = \beta_{n-1}p(t,n-1)C_{\mathrm{vac}} - (\beta_n C_{\mathrm{vac}}+\alpha_n)\,p(t,n) + \alpha_{n+1}p(t,n+1), \tag{13}$$

with initial conditions

$$p(0,n)=1, \quad \text{and} \quad p(0,i)=0, \quad i\neq n, \tag{14}$$

for a population initially of size n. There exist many algorithms that allow to compute an approximation of p at each time, assuming the concentration C_{vac} is known. In the following, we implement a simple but rather efficient algorithm in which particles are propagated according to the law of the Jump process (12). Each particle represents a cluster of size n and evolves independently on a time step δt (during which C_{vac} is constant). Due to the independence of each particle, the method is highly parallelizable. Moreover, as we only consider large size clusters, we do not suffer from high frequency events due to the small clusters behaviour. This will be later illustrated in Section 6.2, where the characteristic jump time appears to be really small for small size clusters.

3.1.2. Jump algorithm: Step (M2)

The Jump process approach interprets the set of rate equations as a set of forward-Kolmogorov equations and its solution is therefore a probability distribution, denoted by $(p(t,n))_{n\geq 2}$ such that $p(t,n)\geq 0$ and $\sum_{n=2}^{\infty} p(t,n)=1$. The total concentration

$$M_{\text{tot}} = \sum_{n=2}^{\infty} \mathscr{C}_{\mathscr{L}}(n) \tag{15}$$

should be stored in order to rescale the probability p and get the concentration as $\mathscr{C}_{\mathscr{L}} = M_{\text{tot}} p$.

In order to compute the law $p(t,n)$, starting from a distribution p_0, the method we propose generates a large number N_{part} of particles $(X_n)_{1\leq n\leq N_{\text{part}}}$ sampled according to p_0. There exist various methods to sample from a multinomial distribution (p_0 is discrete), see for instance [21,22]. These particles are then propagated according to the jump process associated with the transition rates (12). For a particle in state n, its jump frequency is given by

$$\nu(n) = \beta_n C_{\text{vac}} + \alpha_n \tag{16}$$

(*i.e.* the time of the next jump follows an exponential distribution $\mathcal{E}$ of rate $\nu(n)$) and, when it jumps, the particle reaches either the state $n-1$ with probability $\alpha_n/(\beta_n C_{\text{vac}}+\alpha_n)$ or the state $n+1$ with probability $\beta_n C_{\text{vac}}/(\beta_n C_{\text{vac}}+\alpha_n)$. We denote by $\xi(x,\tau,u)$ the function which gives the new state as a function of the previous one x and the two random numbers used in the procedure. Here, τ is a random time sampled from an exponential distribution of parameter $\nu(x)$ and u is a random number sampled from a uniform distribution U on $[0,1]$ allowing us to choose between the state $n-1$ and $n+1$.

There is in fact no notion of fixed time step in this algorithm apart from the time interval δt that corresponds to the final time of the stochastic process. The algorithm summarized as $\mathcal{G}^{\text{large}}_{\delta t}(\mathscr{C}^k_{\mathscr{L}}; C^k_{\text{vac}})$ in equation (M2) at a step k reads as follows:

(1) Sample N_{part} particles according to the initial distribution $p_0^k(n) = M_{\text{tot}}^{-1}\mathscr{C}^k_{\mathscr{L}}(n)$ as:

$$\left(x_1^0, \cdots, x^0_{N\text{part}}\right) \sim p_0^k; \tag{B1}$$

(2) Propagate the particles until $(k+1)\delta t$:
  (a) Associate the ℓ-th particle (denoted by x_ℓ) with a time τ_ℓ^k that is initially set to $k\delta t$;
  (b) Propagate independently all particles:
    (i) first sample a jump time:

$$\begin{aligned}&\text{Compute the jump frequency: } \nu(x_\ell) = \beta_{x_\ell} C^k_{\text{vac}} + \alpha_{x_\ell};\\ &\text{Sample a jump time } \delta\tau_\ell \sim \mathcal{E}(\nu(x_\ell));\\ &\text{Update time } \tau_\ell^k \longleftarrow \tau_\ell^k + \delta\tau_\ell;\end{aligned} \tag{B2.a}$$

    (ii) and then propagate until $(k+1)\delta t$:

$$\begin{aligned}&\text{While } \tau_\ell^k \leq (k+1)\delta t, \text{ do:}\\ &-\text{Sample: } u\sim U;\\ &-\text{Propagate: } x_\ell \longleftarrow \xi(x_\ell, \delta\tau_\ell, u);\\ &-\text{Compute: } \nu(x_\ell) = \beta_{x_\ell} C^k_{\text{vac}} + \alpha_{x_\ell};\\ &-\text{Sample the next jump time } \delta\tau_\ell \sim \mathcal{E}(\nu(x_\ell));\\ &-\text{Update the time as } \tau_\ell^k \longleftarrow \tau_\ell^k + \delta\tau_\ell;\end{aligned} \tag{B2.b}$$

(3) Compute the concentration $\widetilde{\mathscr{C}}^{k+1}_{\mathscr{L}}$ at time $(k+1)\delta t$:

$$\widetilde{\mathscr{C}}^{k+1}_{\mathscr{L}}(n) = \frac{M_{\text{tot}}}{N_{\text{part}}} \sum_{\ell=1}^{N_{\text{part}}} \mathbb{1}_n(x_\ell), \tag{B3}$$

for $n = N_{\text{front}} - N_{\text{buff}}, \cdots, N_{\text{max}}$, and with $\mathbb{1}_i(j) = 1$ if $i=j$ and 0 otherwise.

Notice that at a step $k > 0$ of the main algorithm, Step (B1) can be performed without actually resampling N_{part} particles. To this end, from the previous step $k-1$, one just needs to add the particles coming from the part of the distribution of small size clusters that spills out in the large size clusters zone and delete the ones that are added to the small size cluster distribution. In order to keep the number of particles constant, we either randomly suppress some of the particles if there are more than N_{part} particles, or duplicate some of them if there are less than N_{part} particles.

Such an algorithm differs from the standard SSA procedure in that all particles are handled independently (which itself comes from the fact that the forward-Kolmogorov equation on p is linear). Parallelizing the scheme is straightforward in the present situation and results in a significant improvement in term of wall-clock time. Moreover this method becomes exact in the limit $N_{\text{part}} \to +\infty$. Nevertheless the frequency $\nu(n)$ increases with n and this might reduce the efficiency of the method for very large clusters. In these situations, Fokker–Planck based methods should be used instead.

3.2. The Langevin process approach

3.2.1. Model presentation

Assuming that the size n of the cluster is large enough and that the concentration C_{vac} is known at each time, the rate equations can be approximated with a good approximation by a single Fokker–Planck equation [9,20]:

$$\frac{\partial \mathscr{C}}{\partial t} = -\frac{\partial (F\mathscr{C})}{\partial x} + \frac{1}{2}\frac{\partial^2 (D\mathscr{C})}{\partial x^2}, \tag{17}$$

where $F(t,x) = \beta(x)C_{\text{vac}}(t) - \alpha(x)$ and $D(t,x) = \beta(x)C_{\text{vac}}(t) + \alpha(x)$. The size x of the cluster now plays the role of a spatial coordinate. The scalar field $\mathscr{C}$ acts as a concentration which is continuous in space, with:

$$C_n(t) \simeq \mathscr{C}(t,n) \quad \text{for} \quad n \gg 1. \tag{18}$$

When only one type of defect is considered, such a partial differential equation (PDE) is one-dimensional in size space. In this situation, there exist good solvers to efficiently simulate such equations on large scales problems. Jourdan et al. [10] have proposed an efficient method (based on a finite volume formulation) to numerically solve the Fokker–Planck equation when it is coupled to rate equations.

Nevertheless, when two or more types of defects are introduced, *i.e.* when a cluster is identified by a m-tuple $(n_1, n_2, \cdots, n_m)$, the Fokker–Planck equation is m-dimensional. It then becomes computationally prohibitive to solve it with deterministic mesh-based methods due to the curse of dimensionality (the number of discretization unknowns grows exponentially with the dimension). Stochastic methods are much more appropriate in such situations. The Fokker–Planck equation is related to a stochastic differential equation, called Langevin dynamics. Let $(X_t)_{t\geq 0}$ be the stochastic process

$$dX_t = F(t, X_t)dt + \sqrt{D(t, X_t)}dW_t, \tag{19}$$

where W_t is a standard Wiener process. Then the law $p(t,x)$ of X_t satisfies the Fokker–Planck equation (17). Therefore, by simulating a large number of trajectories for the process X_t, one can obtain a good approximation of the law p, *i.e.* the solution of the Fokker–Planck equation (17). Since the trajectories are independent of each other, this stochastic method is also highly parallelizable.

3.2.2. Langevin process algorithm: Step (M2)

It is important to note that using the stochastic representation of the Fokker–Planck equation (17) allows to evolve a probability $p(t,x)$ such that $\int p(t,x)dx = 1$. As in the Jump process approach, the concentration $M_{\mathscr{L}}$ of large size clusters (15) should therefore be stored in order to rescale the probability p and obtain the concentration $\mathscr{C}_{\mathscr{L}}$.

Let us introduce an interpolation operator $\mathcal{I}$ that transforms a discrete distribution into a continuous one (such as a linear interpolation of the values at integers). The given initial condition $\mathscr{C}^0_{\mathscr{L}}$ is associated with an initial density

$$p_0(x) = \frac{\mathcal{I}\left(\mathscr{C}^0_{\mathscr{L}}\right)(x)}{\int_{N_{\text{front}}}^{\infty} \mathcal{I}\left(\mathscr{C}^0_{\mathscr{L}}\right)(y)dy}, \tag{20}$$

and with a fixed C_{vac}, we formulate the problem as

$$\begin{cases} \text{Find the law } p(t,x) \text{ of } (X_t)_{t\geq 0} \text{ solution of} \\ dX_t = F(X_t)dt + \sqrt{D(X_t)}dW_t, \\ p(0,x) = p_0(x), \end{cases} \tag{21}$$

where F, D are defined after (17). In order to approximate the law $p(t,x)$ one can generate a large number of trajectories for X and use various methods such as histograms or kernel density estimators to construct an empirical density. Let us therefore introduce the number N_{part} of Langevin trajectories, χ a kernel function (*i.e.* a non-negative function that integrates to one and has mean zero) and h a smoothing parameter. We will in particular need to sample the initial distribution p_0 with N_{part} Langevin particles. Here we use a Metropolis algorithm [23,24], starting from $N_{\text{front}} + N_{\text{buff}}$, with a uniform proposal distribution of support $[-\alpha, \alpha]$, with α chosen such that the acceptance ratio is around 0.5. Using the kernel density approach, the law of X_t is approximated by

$$p(t,x) = \frac{1}{N_{\text{part}} h} \sum_{\ell=1}^{N_{\text{part}}} \chi\left(\frac{X_t^\ell - x}{h}\right), \tag{22}$$

where $(X_t^\ell)_{1 \le \ell \le N_{\text{part}}}$ are trajectories of the process X. Finally to propagate the Langevin dynamics, we use a numerical scheme $\psi(x, \Delta t^{\text{L}}, G)$, here a Euler–Maruyama scheme, with a time step Δt^{L}:

$$\psi\left(x, \Delta t^{\text{L}}, G\right) = x + F(x)\Delta t^{\text{L}} + \sqrt{D(x)\Delta t^{\text{L}}}\, G, \tag{23}$$

where G is a standard Gaussian random variable. The time step Δt^{L} is such that $\delta t = K^{\text{L}} \Delta t^{\text{L}}$ for some $K^{\text{L}} \ge 1$.

For the Langevin process approach, the algorithm summarized as $\mathcal{G}_{\delta t}^{\text{large}}(\mathscr{C}_{\mathscr{L}}^k; C_{\text{vac}}^k)$ in equation (M2) at a step k writes:

(1) Sample N_{part} particles according to the initial distribution $p_0^k(x) = \frac{\mathcal{I}\left(\mathscr{C}_{\mathscr{L}}^k\right)(x)}{\int_{N_{\text{front}}}^{\infty} \mathcal{I}\left(\mathscr{C}_{\mathscr{L}}^k\right)(y)dy}$ as:

$$\left(x_1^0, \cdots, x_{N_{\text{part}}}^0\right) \sim p_0^k; \tag{L1}$$

(2) Propagate in time the Langevin particles for $j = 0, \cdots, K^{\text{L}} - 1$:

$$\left(x_1^{j+1}, \cdots, x_{N_{\text{part}}}^{j+1}\right) = \left(\psi\left(x_1^j, \Delta t^{\text{L}}, G_1^j\right), \cdots, \psi\left(x_{N_{\text{part}}}^j, \Delta t^{\text{L}}, G_{N_{\text{part}}}^j\right)\right); \tag{L2}$$

(3) Compute the concentration $\widetilde{\mathscr{C}}_{\mathscr{L}}^{k+1}$:

$$\widetilde{\mathscr{C}}_{\mathscr{L}}^{k+1}(n) = \frac{M_{\text{tot}}}{N_{\text{part}} h} \sum_{\ell=1}^{N_{\text{part}}} \chi\left(\frac{x_\ell^{N_L} - n}{h}\right), \tag{L3}$$

for $n = N_{\text{front}} - N_{\text{buff}}, \cdots, N_{\max}$.

Once again the stochastic particles are independent and a parallelization of the method is straightforward. Moreover the same remark as in Section 3.1.2 holds for Step (L1): we similarly avoid a full resampling and simply update the population size once the distributions have been separated again into distributions of small and large clusters.

4. Approximating the dynamics of C_{vac}

The value of C_{vac} needs to be calculated at multiple values of the time increment δt in Step (M4) of the main algorithm. This is encoded in the numerical method $C_{\text{vac}}^k = \mathcal{F}(C_{\text{vac}}^{k-1}; C^k)$. We present here three methods to this end.

4.1. Decomposition into elementary integrable ODEs

The first method consists in solving the ODE (9) with fixed concentrations $(C_n)_{n \ge 2}$. A direct integration of the ODE with standard schemes is not appropriate in our case because the values $\mathcal{A}^k$ and $\mathcal{B}^k$ are fluctuating every time step δt due to the stochastic evolution of large size clusters. Since the right hand side term of Eq. (9) is the difference of two large terms and is observed to be small when integrating the full cluster dynamics (see Fig. 6), small fluctuations create large instabilities. In order to develop a stable numerical scheme we recommend a decomposition of the evolution into two integrable parts (an affine part and a nonlinear one). For fixed $\mathcal{A}^k$ and $\mathcal{B}^k$ (defined in Eq. (7) and (8)) we split Eq. (9) into the affine part

$$\frac{dC_{\text{vac}}^{\text{L}}}{dt} = -\mathcal{B}^k C_{\text{vac}}^{\text{L}} + \mathcal{A}^k, \tag{24}$$

and the non-linear one

$$\frac{dC_{\text{vac}}^{\text{NL}}}{dt} = -2\beta_1 \left(C_{\text{vac}}^{\text{NL}}\right)^2. \tag{25}$$

Both equations exhibit analytic solutions in closed forms, namely:

$$C_{\mathrm{vac}}^{\mathrm{L}}(t+t_{\mathrm{init}}) = \left(C_{\mathrm{vac}}^{\mathrm{L}}(t_{\mathrm{init}}) - \frac{\mathcal{A}^k}{\mathcal{B}^k} \right) \exp(-\mathcal{B}^k t) + \frac{\mathcal{A}^k}{\mathcal{B}^k}, \tag{26}$$

$$C_{\mathrm{vac}}^{\mathrm{NL}}(t+t_{\mathrm{init}}) = \frac{C_{\mathrm{vac}}^{\mathrm{NL}}(t_{\mathrm{init}})}{1+2\beta_1 t C_{\mathrm{vac}}^{\mathrm{NL}}(t_{\mathrm{init}})}. \tag{27}$$

To compute the solution one may adopt either a first or second order scheme. To integrate the dynamics with a time step Δt, such that $\delta t = J\Delta t$, the second order scheme writes

$$\begin{cases} C_{\mathrm{vac}}^{k-1,j+1/2} = \left(C_{\mathrm{vac}}^{k,j} - \dfrac{\mathcal{A}^k}{\mathcal{B}^k} \right) \exp\left(-\dfrac{\mathcal{B}^k \Delta t}{2} \right) + \dfrac{\mathcal{A}^k}{\mathcal{B}^k}, \\ \widetilde{C}_{\mathrm{vac}}^{k-1,j+1} = \dfrac{C_{\mathrm{vac}}^{k,j+1/2}}{1+\beta_1 \Delta t C_{\mathrm{vac}}^{k,j+1/2}}, \\ C_{\mathrm{vac}}^{k-1,j+1} = \left(\widetilde{C}_{\mathrm{vac}}^{k,j+1} - \dfrac{\mathcal{A}^k}{\mathcal{B}^k} \right) \exp\left(-\dfrac{\mathcal{B}^k \Delta t}{2} \right) + \dfrac{\mathcal{A}^k}{\mathcal{B}^k}. \end{cases} \tag{28}$$

The value C_{vac}^k is set to $C_{\mathrm{vac}}^{k-1,J}$ in Step (M4) of the main algorithm.

4.2. Quasi-stationary limit

The second method consists in making a stronger assumption on the behaviour of C_{vac}, namely that

$$\frac{dC_{\mathrm{vac}}}{dt} \simeq 0. \tag{29}$$

This situation occurs in many physical systems and the quasi-stationary limit is a good approximation in many cases [15]. The vacancy concentration is then given by the positive solution of the second order equation

$$-2\beta_1 (C_{\mathrm{vac}})^2 - \mathcal{B}^k C_{\mathrm{vac}} + \mathcal{A}^k = 0. \tag{30}$$

The two solutions of this equation are

$$r_\pm = -\frac{\mathcal{B}^k}{4\beta_1} \pm \frac{\sqrt{\left(\mathcal{B}^k\right)^2 + 8\beta_1 \mathcal{A}^k}}{4\beta_1}. \tag{31}$$

Since $r_- < 0$, the only physical solution is $r_+ > 0$. The second way of implementing (M4) is therefore given by

$$C_{\mathrm{vac}}^{k+1} = \frac{2\mathcal{A}^k}{\mathcal{B}^k} \left(1 + \sqrt{1 + \frac{8\beta_1 \mathcal{A}^k}{\left(\mathcal{B}^k\right)^2}} \right)^{-1}, \tag{32}$$

which is equal to r_+ in Eq. (31), though numerically more stable when $\mathcal{A}$ and $\mathcal{B}$ are large.

4.3. Mass conservation

The third method is based on the preservation of the total quantity of matter (see Eq. (5)):

$$C_{\mathrm{vac}} + \sum_{n\geq 2} n C_n(t) = C_{\mathrm{init}}.$$

The computation of C_{vac} given the concentrations $(C_n)_{n\geq 2}$ is then straightforward:

$$C_{\mathrm{vac}}^k = C_{\mathrm{init}} - \sum_{n\geq 2} n C_n^k. \tag{33}$$

5. Numerical analysis

We give in this section error estimates for the schemes we consider. We focus our analysis on two sources of errors: systematic errors due to the splitting of the dynamics (see Section 5.1), as well as sampling errors related to the introduction of stochastic algorithms (see Section 5.2). We conclude the section by a discussion on the total error which results from these two sources (see Section 5.3).

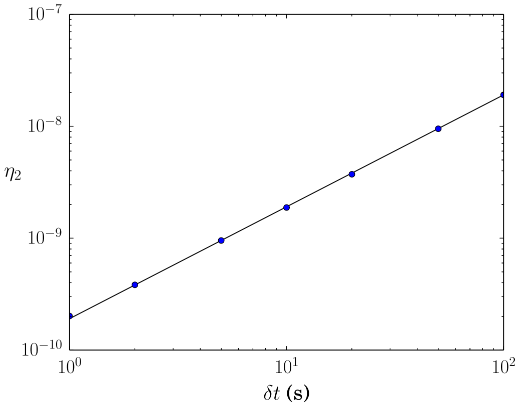

Fig. 4. Error η_2 at final time $t_f = 1000$ s as a function of δt. A linear approximation is superimposed in a black line.

5.1. Deterministic splitting

Since the splitting introduced in Section 2.2 is a Lie–Trotter splitting, we expect to observe an error of order δt for properties computed over finite time intervals (which is a standard result in the numerical analysis of ODEs). In order to illustrate such error estimates, we run simulations with various time steps δt, in the case when the elementary sub-ODEs in (P1) are both integrated using discretization techniques for ODEs in order not to pollute error estimates with sampling errors. The elementary sub-ODEs are integrated in time using standard second order numerical scheme (Heun scheme) when necessary, and with very small time steps Δt, in order to limit errors due to the underlying numerical scheme and retain only the error coming from the splitting algorithm. The solutions for C_{vac} are obtained by using the first method presented in Section 4.1, as the error of the integration scheme scales as Δt^2. A reference solution is obtained by computing a solution C^{ref} to the full ODE, with the same small timestep Δt. The errors between the numerical solutions $C^{\text{splitting}}$ of problem (P1) and the reference solution C^{ref} are measured with the following mean square error norm:

$$\eta_2^{\text{splitting}}(t) = \sqrt{\sum_{n=1}^{N_{\max}} \left(C_n^{\text{splitting}}(t) - C_n^{\text{ref}}(t) \right)^2}. \tag{34}$$

The simulations are performed using the setting described more precisely in Section 6, for a time $t_f = 1000$ s and with $\Delta t = 10^{-5}$ s. In Fig. 4, we compare the error at the final time for various values of δt, ranging from 10^0 s to 10^2 s. As expected, the error η_2 decreases linearly with the splitting time step δt.

5.2. Stochastic error

Another source of error in the algorithm is the sampling error arising from the finiteness of the number of particles N_{part} used to approximate large cluster size dynamics. Since these particles evolve independently, a central limit theorem can be applied, showing that the statistical error scales as $N_{\text{part}}^{-1/2}$. We illustrate such a behaviour by running simulations of a simple system where all clusters are handled stochastically using the jump process (in order to avoid the error coming from the Fokker–Planck approximation), except for the single vacancies, which represent a very small part of the total mass. To measure the error, we consider the average size of clusters, which corresponds to the following observable $\mathcal{M}$ (using the notation of Section 3.1)

$$\mathcal{M}(C) = \frac{\displaystyle\sum_{n\ge1} nC_n}{\displaystyle\sum_{n\ge1} C_n} = \frac{\displaystyle\sum_{n\ge1} n \left(\sum_{\ell=1}^{N_{\text{part}}} \mathbb{1}_n(x_\ell) \right)}{\displaystyle\sum_{n\ge1} \left(\sum_{\ell=1}^{N_{\text{part}}} \mathbb{1}_n(x_\ell) \right)}. \tag{35}$$

For each value of N_{part}, we perform N_{sim} independent simulations, denoted by $(C^j)_{1\le j\le N_{\text{sim}}}$, in order to quantify the variability of the outcome. At a given final time t_f, we expect that, when N_{part} is sufficiently large for a central limit theorem to hold, the empirical variance of the outcome scales as $N_{\text{part}}^{-1/2}$. More precisely, denoting by $C^{t_f,j}$ the concentrations at time t_f for the j-th realization, this empirical variance reads

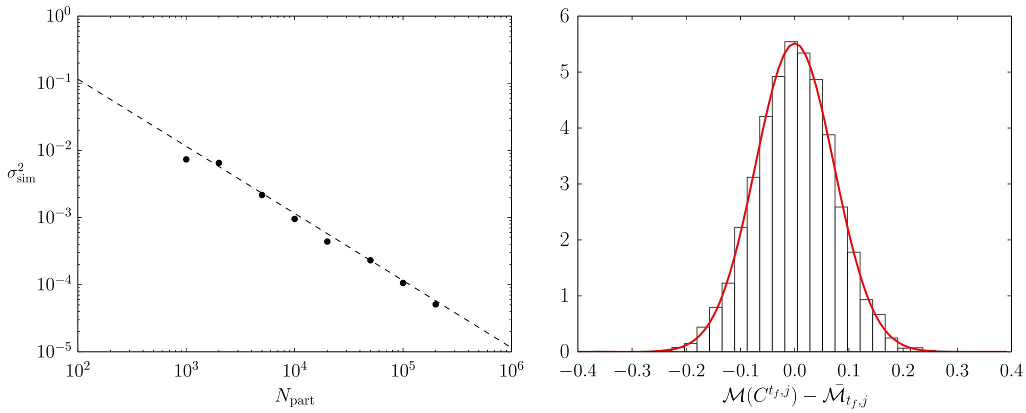

Fig. 5. Left: Variance (36) as a function of $N_{\rm part}$. A linear approximation is superimposed in dashed line. Right: Histogram of the outcomes $\mathcal{M}(C^{t_{\rm f},j}) - \overline{\mathcal{M}}_{t_{\rm f},j}$. The reference Gaussian distribution with variance $\mathcal{N}(0, \sigma^2_{\rm sim})$ is superimposed to the data.

$$\sigma^2_{\rm sim} = \frac{1}{N_{\rm sim}} \sum_{j=1}^{N_{\rm sim}} \left(\mathcal{M}\left(C^{t_{\rm f},j}\right) - \overline{\mathcal{M}}_{t_{\rm f},N_{\rm sim}} \right)^2 \simeq \frac{\sigma^2(t_{\rm f})}{N_{\rm part}}, \tag{36}$$

with

$$\overline{\mathcal{M}}_{t_{\rm f},N_{\rm sim}} = \frac{1}{N_{\rm sim}} \sum_{n=1}^{N_{\rm sim}} \mathcal{M}\left(C^{t_{\rm f},j}\right)$$

the average value of the observable $\mathcal{M}$ over the realizations. There is however no prediction for the asymptotic variance $\sigma^2(t_{\rm f})$, which has to be estimated numerically. In fact, the outcomes $\mathcal{M}(C^{t_{\rm f},j})$ should be distributed according to a Gaussian distribution of variance $\sigma^2(t_{\rm f})/N_{\rm part}$.

The scaling (36) is illustrated in Fig. 5 (left), for $N_{\rm sim} = 100$ and $N_{\rm part}$ ranging from 10^3 to 2×10^5, with final simulation time $t_{\rm f} = 10^3$ s. In Fig. 5 (right), we check the fact that the outcomes are indeed distributed according to a Gaussian distribution, in the case when $N_{\rm part} = 2 \times 10^3$ and $N_{\rm sim} = 10^4$.

5.3. Coupling of errors

We identified in the previous sections two important sources of errors: systematic errors arising from the splitting and sampling errors arising from a discretization of the stochastic representation of the solution. These two sources of errors are the only ones when the integration error in the ODE describing the small size clusters is negligible and jump processes are used for the stochastic parts. The total error at time t on some observable of interest then writes

$$\eta_2^{\rm tot}(t) \simeq \varepsilon_1(t)\delta t + \frac{\varepsilon_2(t)}{\sqrt{N_{\rm part}}}, \tag{37}$$

where $\varepsilon_1(t)$ and $\varepsilon_2(t)$ depend on the simulation time t (as the notation suggests) and only on the chosen observable. In all our simulations, with splitting time steps no larger than $\delta t = 10$ s and various observables such as the error $\eta_2(t)$ at a given time, see Eq. (34), or the total mass, we observe that $\varepsilon_1(t)\delta t \ll \eta_2^{\rm tot}(t)$ for $N_{\rm part}$ ranging from 10^3 to 2×10^7. The statistical error therefore appears to be the main source of error.

6. Results

All the simulations reported below are performed using the parameters of [18], which are summarized in Table 1. The final computation time $t_{\rm f}$ as well as the time interval δt and the time steps $\Delta t^{\rm M}$, $\Delta t^{\rm L}$ (respectively used for the deterministic integration of Eq. (1) and the integration of Langevin dynamics in Step (L2)) are specified for each result. Besides the numerical analysis conducted in Section 5, there is a systematic error due to the Fokker–Planck approximation. This error is expected to be of order $N_{\rm front}^{-2/3}$. A more thorough study of the Fokker–Planck approximation will be given in a future paper since the estimation of this error requires to first provide a mathematically rigorous derivation of the Fokker–Planck limit. Nevertheless, the simulations we performed (not reported here) show that, with $N_{\rm front} = 200$, the error due to the Fokker–Planck approximation is negligible compared to the statistical error as long as the number of particles $N_{\rm part}$ does not exceed 10^8. The buffer zone is of size $2N_{\rm buffer} = 100$ and therefore extends from $n = 150$ to $n = 250$.

Table 1
Parameters for the simulation of a nickel-like metal.

Temperature (T)	823 K
Atomic volume (V_{at})	1.205×10^{-29} m^3
Vacancy formation energy (E^{f}_{vac})	1.7 eV
Vacancy migration energy (E^{m}_{vac})	1.1 eV
Vacancy diffusion coefficient (D_{vac})	$10^{-6}\exp(-E^{m}_{vac}/(k_B T))$ m^2 s^{-1}
Surface energy (γ)	1.0 J/m^2
Concentration of quenched-in vacancies (C_{init})	10^{-7} atom^{-1}
Kernel function (χ)	$(2\pi)^{-1/2}\exp(-x^2/2)$
Smoothing parameter (h)	0.3

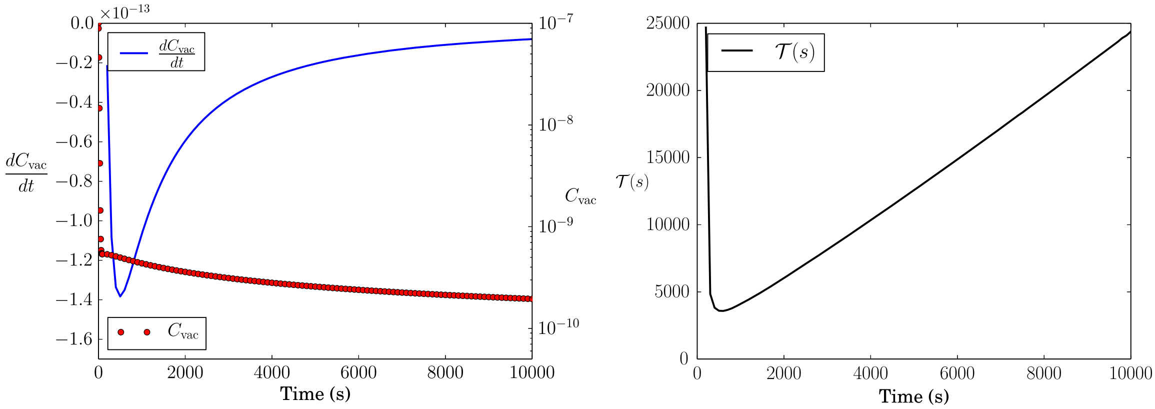

Fig. 6. Left: C_{vac} and its derivative as functions of time. Right: characteristic time $\mathcal{T}$ as a function of time.

6.1. On the quasi-stationary assumption

We first show that under the assumption that C_{vac} reaches a quasi-stationary state, the problem (P1) is an excellent approximation of the original RECD problem (1)–(4) even for large time intervals δt. Let us introduce a time-dependent function $\mathcal{T}$ defined by

$$\mathcal{T}(t) = \left| \frac{1}{C_{vac}(t)} \frac{dC_{vac}}{dt} \right|^{-1}. \tag{38}$$

The function $\mathcal{T}$ acts as a characteristic time for the vacancy density evolution. Since

$$C_{vac}(t+\delta t) = C_{vac}(t) + \frac{dC_{vac}}{dt}(t)\delta t + O(\delta t^2), \tag{39}$$

on a time step $\delta\tau \ll \mathcal{T}$, the variation of C_{vac} is relatively small. The condition $\delta t \ll \mathcal{T}$ indicates the relevant orders of magnitude of δt.

We compute a reference solution by solving the full ODE problem (1)–(4) without any approximation other than the integration scheme. We use a second order Euler–Heun numerical scheme. Starting from the initial condition (6), the system first goes through a nucleation stage (for which we use a small time step $\Delta t^{M} = 10^{-5}$ s), before it enters a growth regime where we start to observe the quasi-stationary state of C_{vac} (characterized by $\frac{dC_{vac}}{dt} \simeq 0$). This state is rapidly observed and is reached before clusters grow beyond N_{front}. After the nucleation stage as the system enters the growth regime, we use the time step $\Delta t^{M} = 10^{-3}$ s. This time step ensures a good conservation of the total quantity of matter $\mathcal{Q}_{tot} = C_{vac} + \sum_{n=2}^{N_{max}} nC(n)$, with a relative error of order 10^{-7}. The final time of computation t_f is set to 10^4 s. We denote by C^{ref} the numerical solution of the dynamics (1)–(4) obtained by this procedure. The solution C^{ref} allows to compute the characteristic time $\mathcal{T}$ defined in Eq. (38). We observe in Fig. 6 that C_{vac} strongly decreases at first from its initial value $C_{init} = 10^{-7}$ and then enters a quasi-stationary state where the concentration of vacancy slowly decreases. At time $t = 10^3$ s, the characteristic time $\mathcal{T}(t)$ is approximately equal to 4×10^3 s. This indicates that a time step δt of order 10^2 s is sufficiently small in order to keep the variations of C_{vac} small.

We next compare the three methods of computing C_{vac}, using a Euler–Heun integration of problem (P1) to update the cluster concentrations (still with $\Delta t^{M} = 10^{-3}$ s). The initial condition is set to $C^{ref}(t_0)$ and $t_0 = 10^3$ s ensures that the initial transient regime is over. We then compute the solution obtained when updating the vacancy concentration every time step

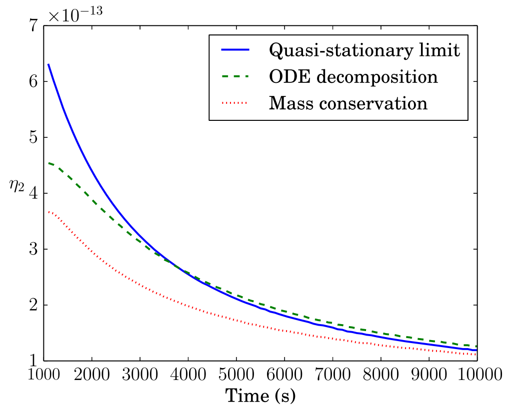

Fig. 7. Behaviour of the mean square error (34) as a function of δt (Quasi-stationary limit: Eq. (32); ODE decomposition: Eq. (28); Mass conservation: Eq. (33)).

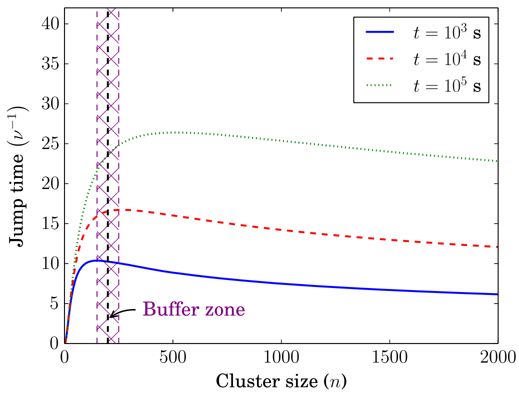

Fig. 8. Characteristic jump time ν^{-1} (see Eq. (16)) as a function of the cluster size n: the most frequent events occur for small n.

$\delta t = 10$ s using one of the three methods discussed in Section 4, until $t_f = 10^4$ s. For the method presented in Section 4.1, we use $\Delta t = 10^{-3}$ s. We then estimate the mean square error between the full ODE solution C^{ref} and the solutions $C^{\text{splitting}}$ of problem (P1) as defined by (34). Fig. 7 compares the errors for each of the three ways of integrating the dynamics of C_{vac} for $C^{\text{splitting}}$. It shows that there is no significant differences between the three methods, as the error remains more than 5 orders of magnitude lower than the total quantity of matter.

In the sequel, we choose the quasi-stationary limit approximation (32) as it is stable and straightforward to compute. The conclusion could be different when other types of mobile clusters are taken into account (typically small clusters such as $C_2, \cdots, C_{10}$). In this case the mass conservation method cannot be used. The decomposition into elementary integrable ODEs then becomes the best alternative since the quasi-stationary limit approach requires solving a system of coupled non-linear equations.

6.2. Accuracy of the splitting algorithm for thermal ageing

We now present simulation results obtained with the main algorithm presented in Section 2.3, using the methods presented in Section 3.1 and 3.2 for the large cluster dynamics. In Steps (B2.a)–(B2.b) and (L2) of the large size cluster dynamics, each particle is propagated independently, which allows dispatching the computations on a parallel architecture. The computations reported here were performed on a cluster of 15 hyperthreaded cores, each thread being used to propagate $N_{\text{proc}} = 2.10^5$ particles, which gives us a total of $N_{\text{part}} = 6.10^6$ particles. The final time of computation t_f is set to 10^5 s.

The time step used in Langevin process simulations is set to $\Delta t^{\text{L}} = 1$ s while the concentration of vacancies C_{vac} is updated[1] at times that are multiple of $\delta t = 10$ s. The value of C_{vac} is calculated using the quasi-stationary limit approach (32). For the Jump approach, the time step is not fixed but a characteristic jump time is given by the particles of size N_{front} and is on the order $(\beta_{N_{\text{front}}} C_{\text{vac}} + \alpha_{N_{\text{front}}})^{-1}$ which is on the order of 10 s when the system is in a growth regime (see Fig. 8).

[1] We could have chosen larger time steps and time intervals in order to speed up the computational time. However we refrained from optimizing the parameters and comparing with state of the art methods since our main objective is to use our method in more complex problems than the ones which can be currently solved with classical methods.

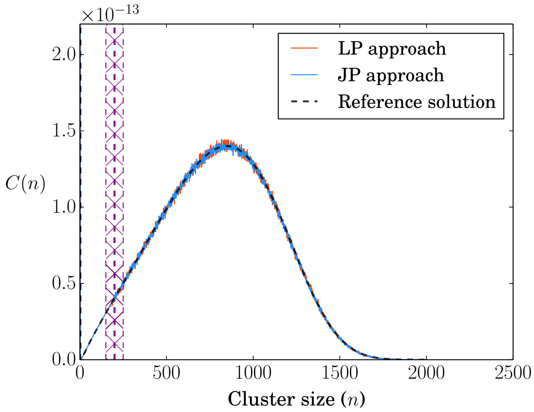

Fig. 9. Comparison between the exact concentration (black dotted line), the concentration obtained with a Langevin process (LP) approach (red line) and the one obtained with the Jump process (JP) approach (blue line). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

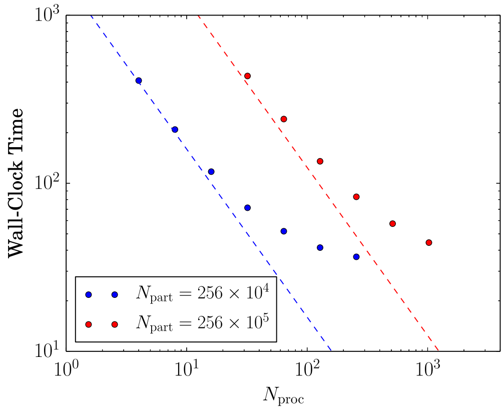

Fig. 10. Comparison between the wall-clock time of computations for a fixed number of particles N_{part}. The ideal linear scaling behaviour is plotted in dashed lines.

Moreover it is increasing with time. In contrast, with a fully stochastic approach (without the decomposition between small and large size clusters), the time steps of the most frequent events are of order $(\beta_2 C_{\text{vac}} + \alpha_2)^{-1} \simeq 10^{-2}$ s.

Aside from the stochastic fluctuations inherent to both methods, the results presented in Fig. 9 show a perfect agreement with the exact concentration obtained by an integration of the full ODE system. The total concentration is equal to $\mathcal{Q}_{\text{tot}}^{\text{LP}} = 9.989 \times 10^{-8}$ at the final time in the case of the LP approach and is equal to $\mathcal{Q}_{\text{tot}}^{\text{JP}} = 9.968 \times 10^{-8}$ for the JP approach. The relative error on the total concentration is therefore less than 0.4%.

6.3. Parallelization

We finally investigate the gain in efficiency arising from the parallelization of the stochastic part, a crucial advantage of the proposed method. We assess relative efficiencies by computing and comparing the wall-clock time of our simulations depending on the number of processors we used to run the simulations. We report in Fig. 10 the results of simulations performed with fixed number of particles ($N_{\text{part}} = 256 \times 10^4$ or $N_{\text{part}} = 256 \times 10^5$) and various numbers of processors N_{proc} ranging from 4 to 1024. We observe that the increase of the number of processors at first reduces linearly the wall-clock computation time, but then, as the number of particles per processor decreases, we observe that the slope tends to reach an asymptotic limit, determined by computational time associated with the deterministic part. Moreover we notice that, as expected, this asymptotic limit appears for smaller values of N_{proc} when the total number of stochastic particles decreases.

7. Conclusion

In this article we have presented a generic coupling algorithm allowing to simulate thermal ageing and ageing under irradiation using cluster dynamics. Our approach consists in coupling the standard rate equations for small size clusters with more efficient stochastic methods for large size clusters. Such a coupling is based on a splitting of the dynamics between the nonlinear dynamics of the vacancy concentration and the linear evolution of the cluster concentrations at fixed vacancy concentration. The dynamics of cluster concentrations is integrated by decomposing the initial condition and independently evolving the dynamics of small and large clusters.

We emphasized two stochastic methods in order to simulate the evolution of the concentration of large size clusters. The Fokker–Planck approximation is well known, but our stochastic treatment with Langevin dynamics is more recent [15] in the cluster dynamics community. The Jump process approach is reminiscent of event kinetic Monte Carlo algorithms [11] but can be parallelized much more efficiently and high frequency events associated with small size clusters are avoided. The main interest of these approaches is that they can be extended to higher dimensional situations. Moreover, with both methods, the particles are propagated independently, which allows to dispatch computations on a parallel architecture, henceforth decreasing the wall-clock computation time. With both methods the quantity of matter is accurately conserved and the distribution of concentrations we obtain is very close to the exact solution obtained by a numerical integration of the original full ODE system.

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